VOICE CONTROLLED ROBOT

Noor ul hudda memon(15SW152)

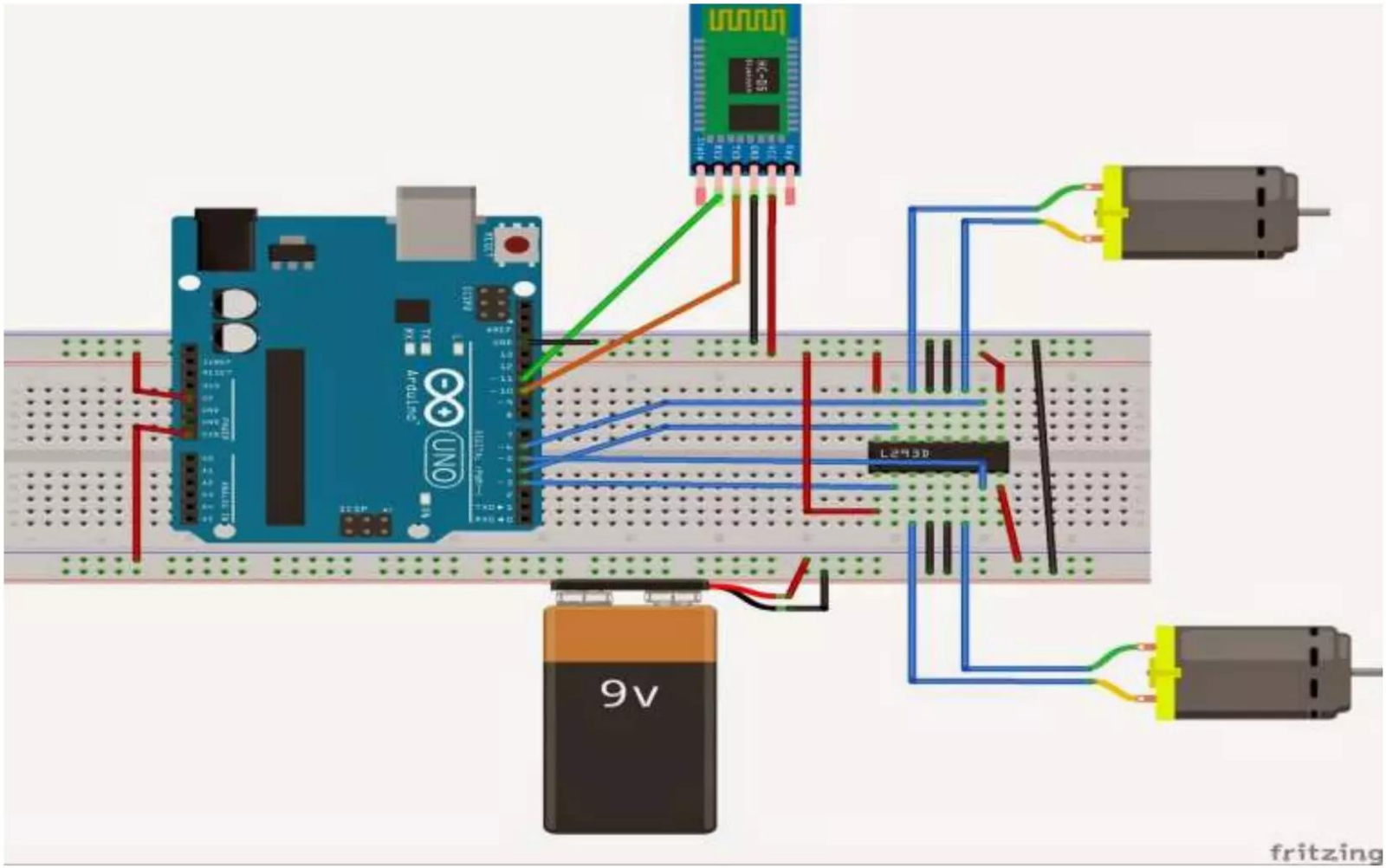
Overview

- Introduction
- Block Diagram
- Hardware Requirements
- Software used
- Applications

INTRODUCTION

This project proposes the use of an android mobile to control a robot via voice commands.

BlockDiagram



Required Components:

- **Arduino Uno**



- **Bluetooth Module (**HC-05)



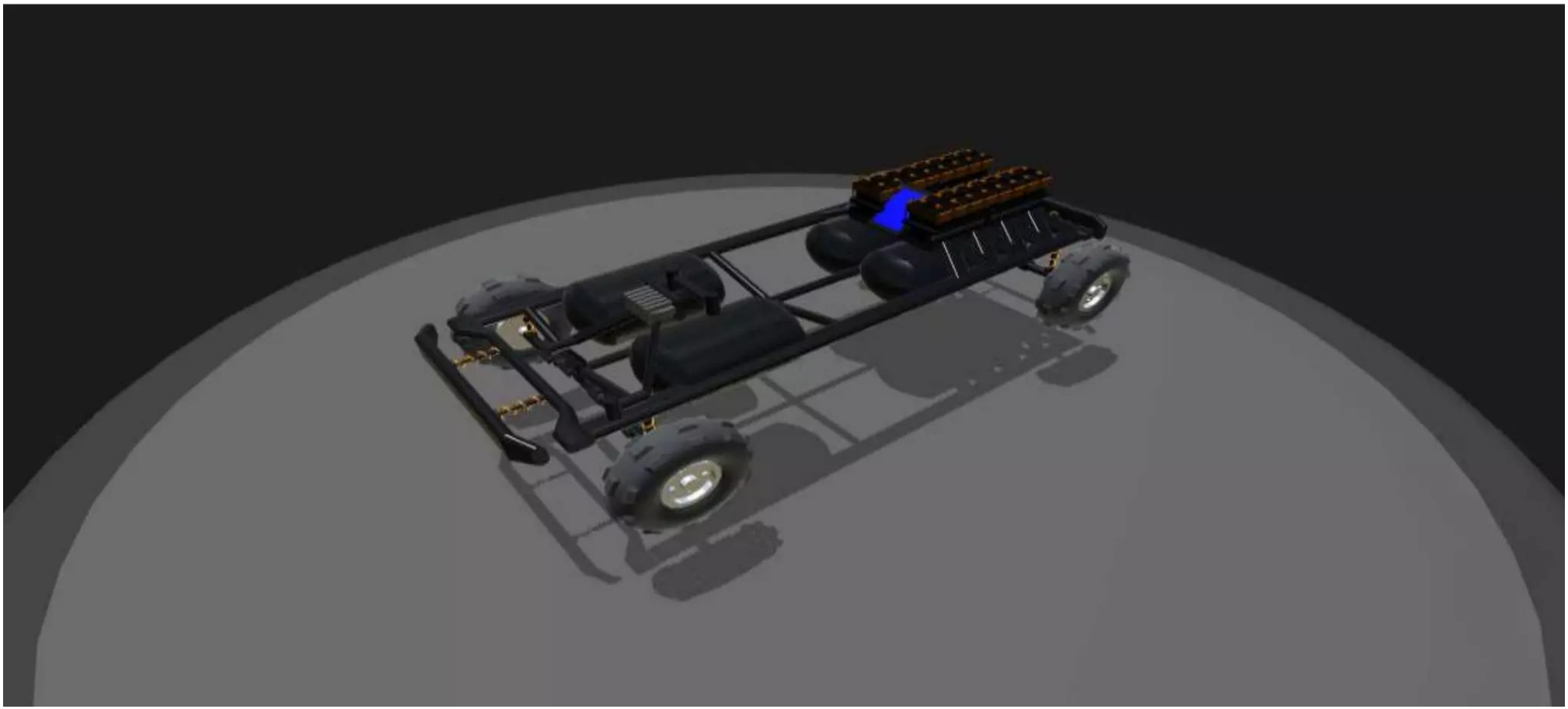
- **L293D Motor Driver**



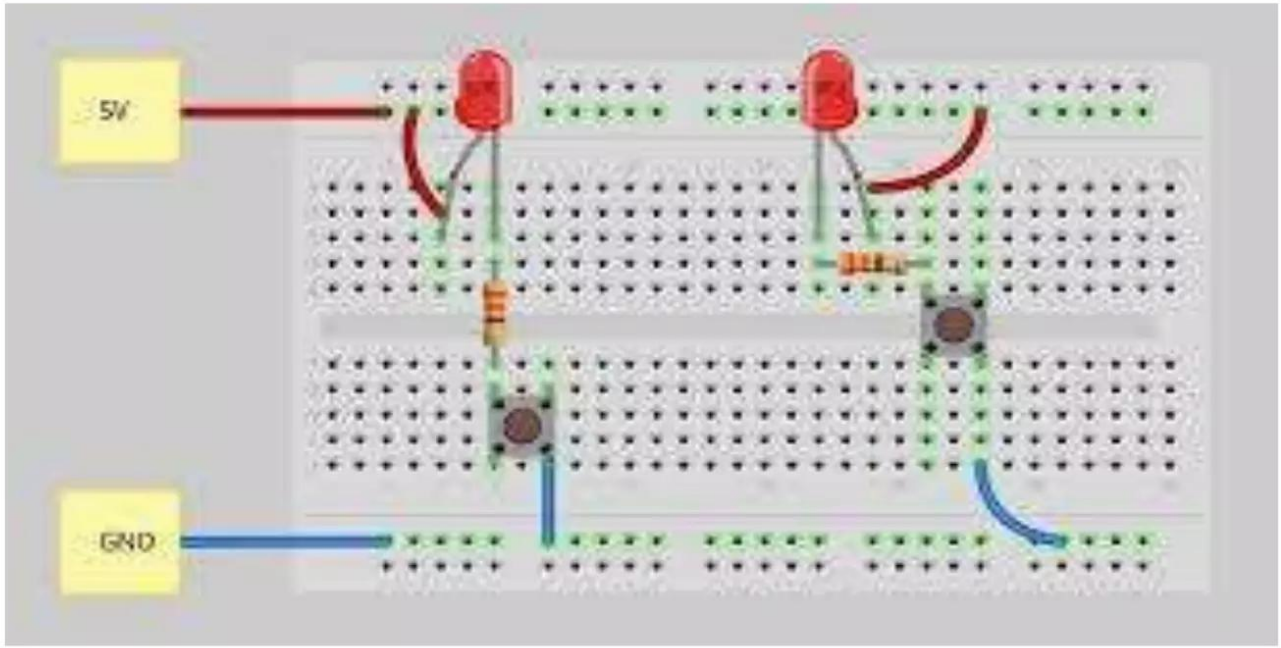
- **Gear Motor and Wheel**



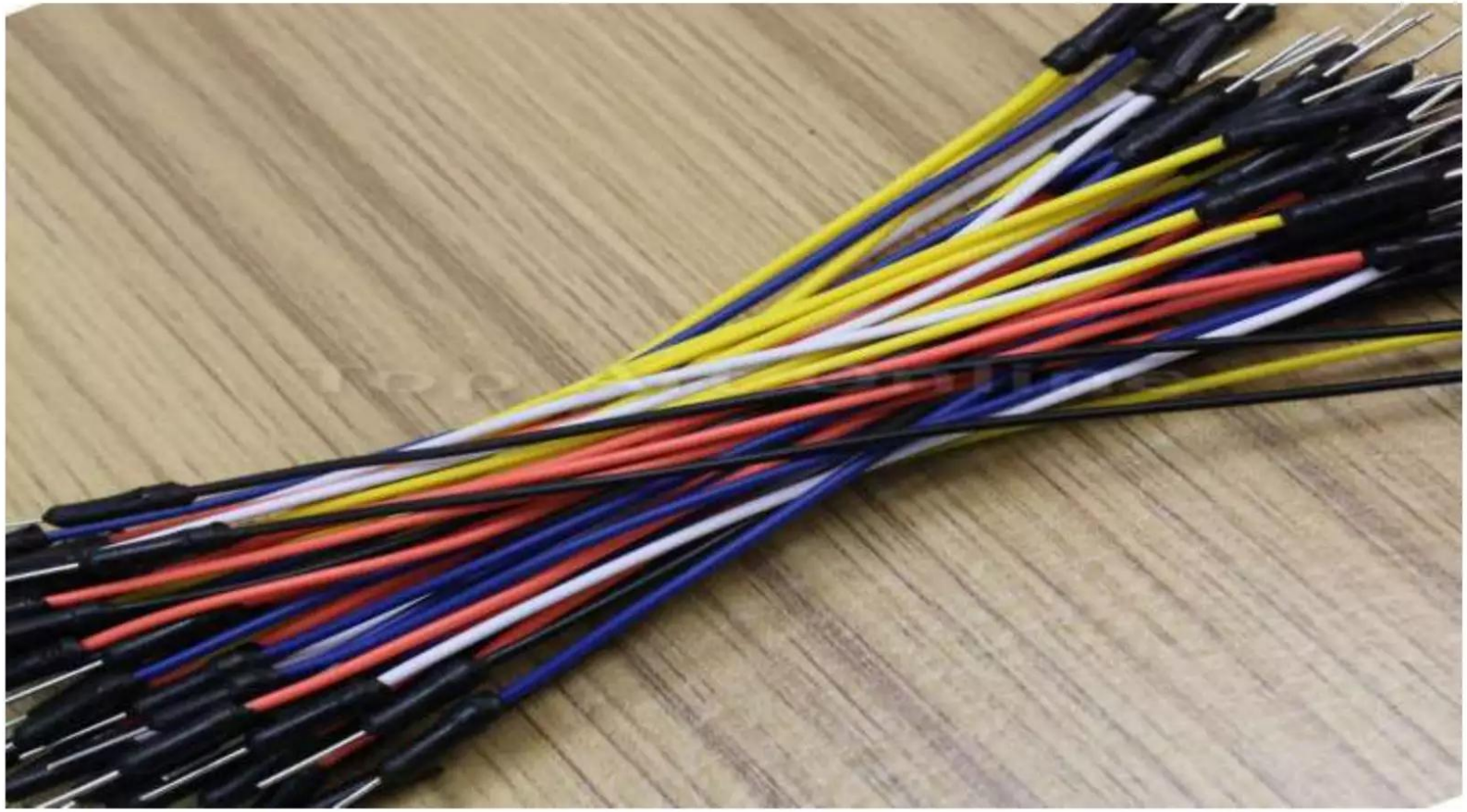
- **Chassis**



- **A breadboard**



- **Connecting wires**



Pin Diagram

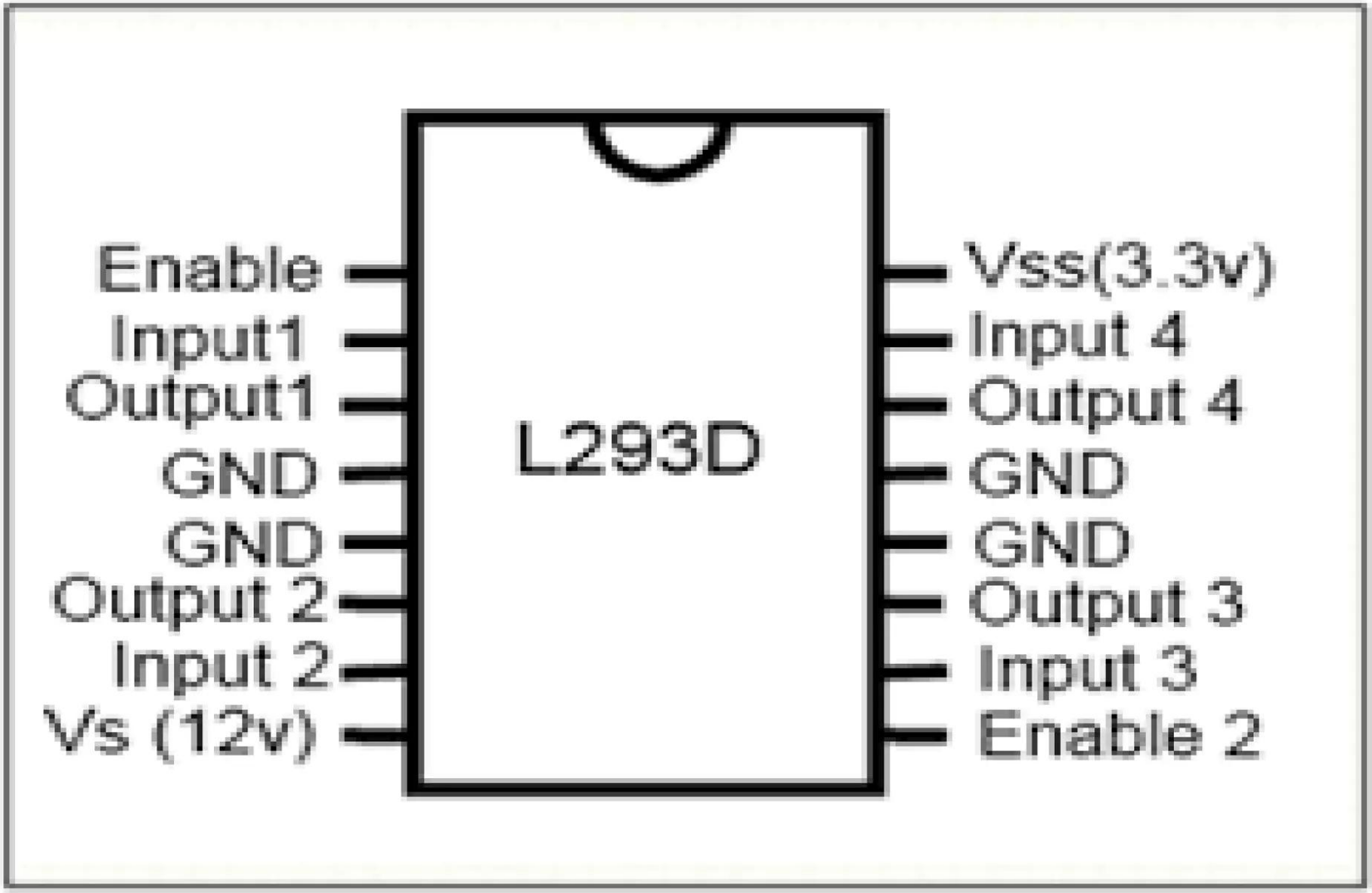
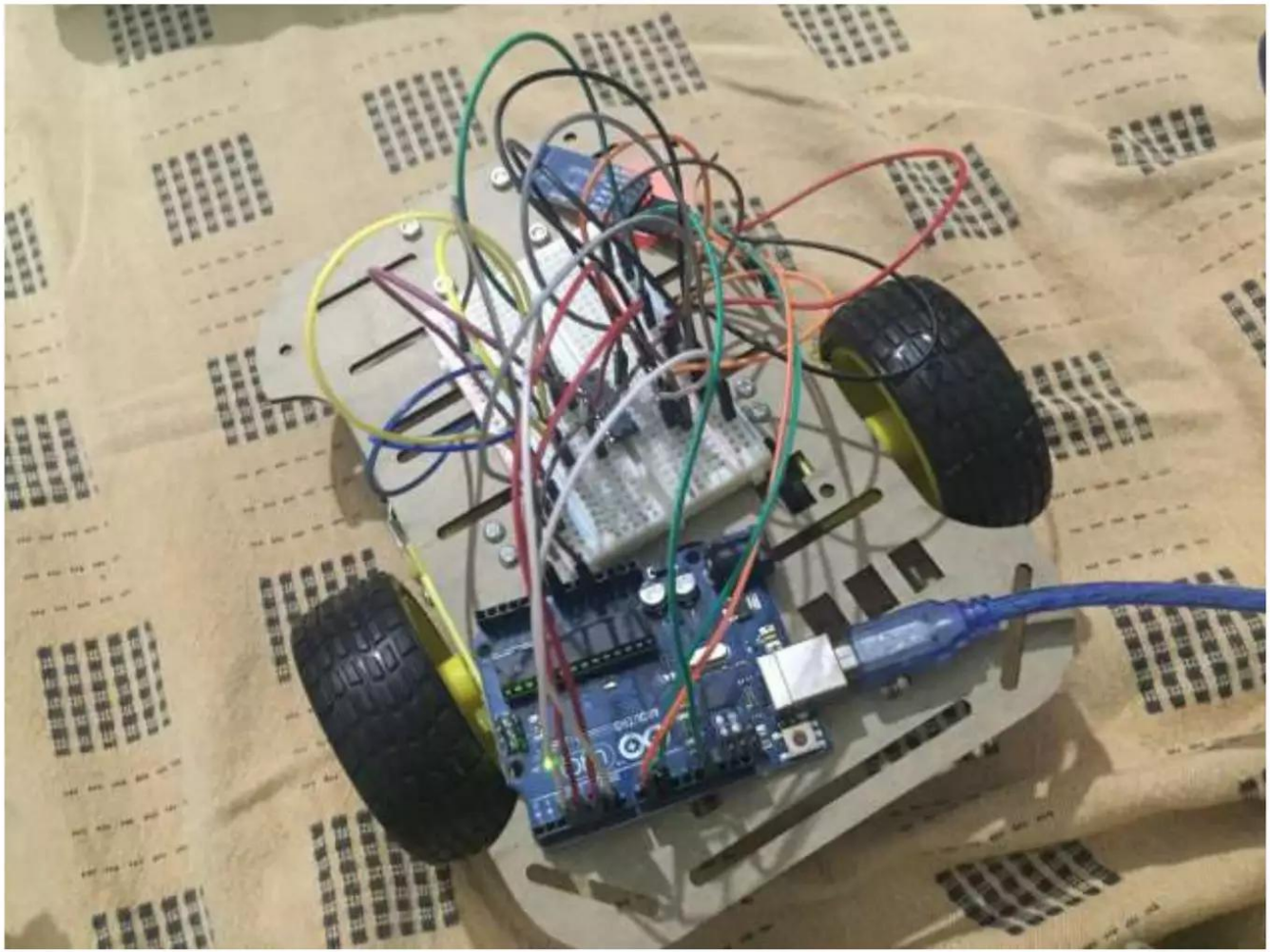

Enable
Input1
Output1
GND
GND
Output 2
Input 2
Vs (12v)
L293D
Vss(3.3v)
Input 4
Output 4
GND
GND
Output 3
Input 3
Enable 2

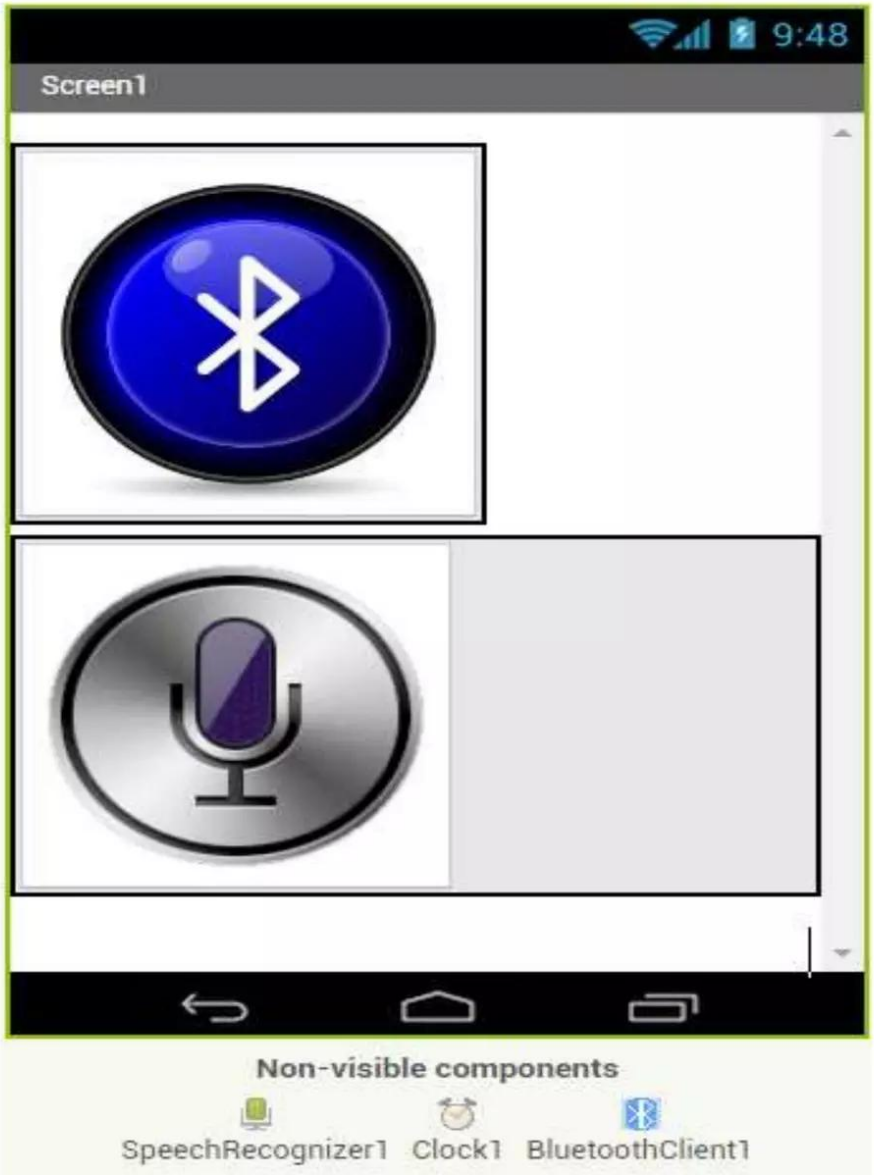
Table-L293D

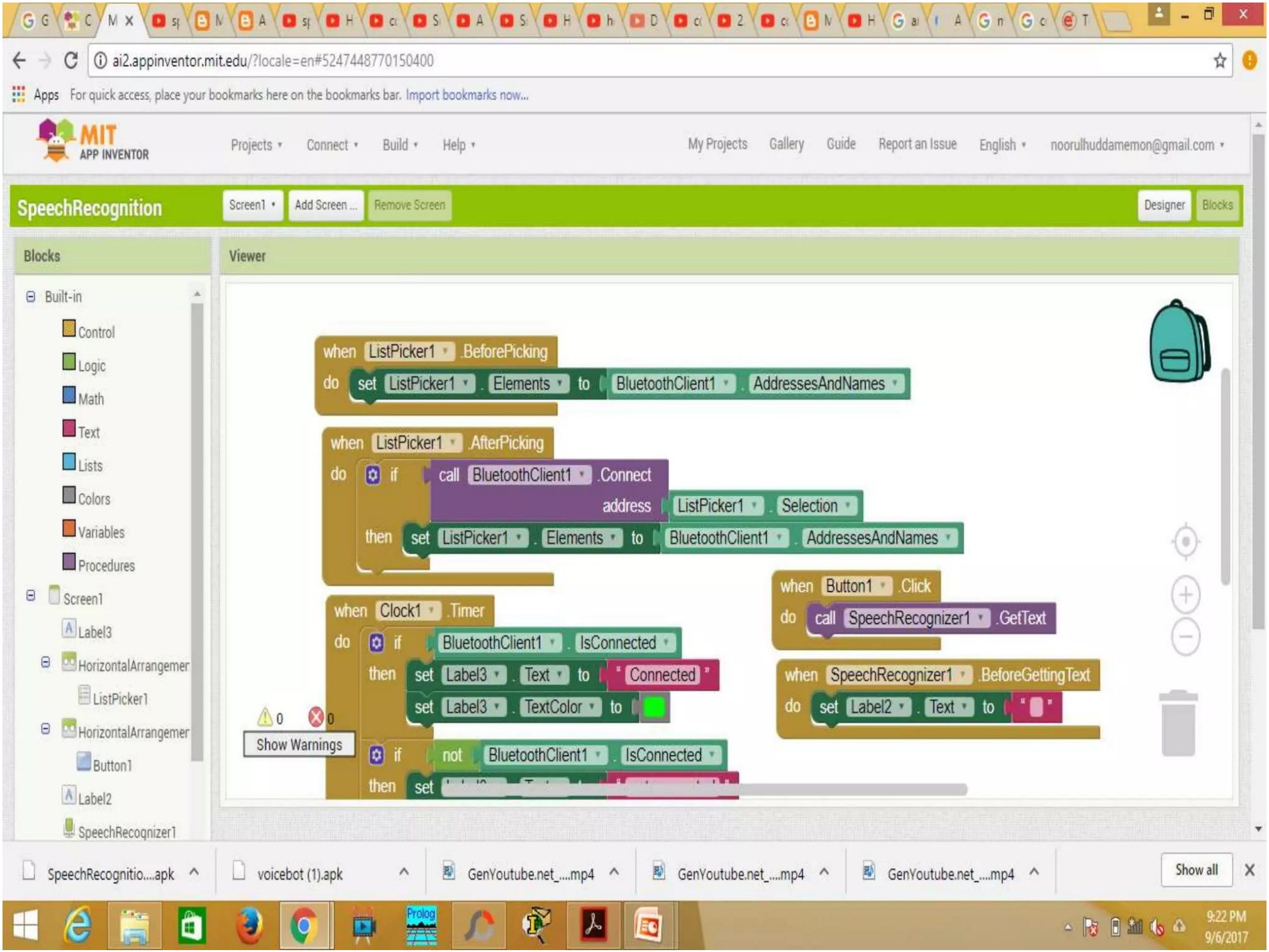
L293D Inputs				Motors	
Input 1	Input 2	Input 3	Input 4	Motor 1	Motor 2
1	0	1	0	Fwd	Fwd
1	0	1	0	Fwd	Fwd
1	0	0	1	Fwd	Back
1	0	0	0	Fwd	Stop
0	1	1	0	Back	Fwd
0	0	1	0	Stop	Fwd
0	1	0	1	Back	Back
0	0	0	0	Stop	Stop

Voice Recognition Robot Car

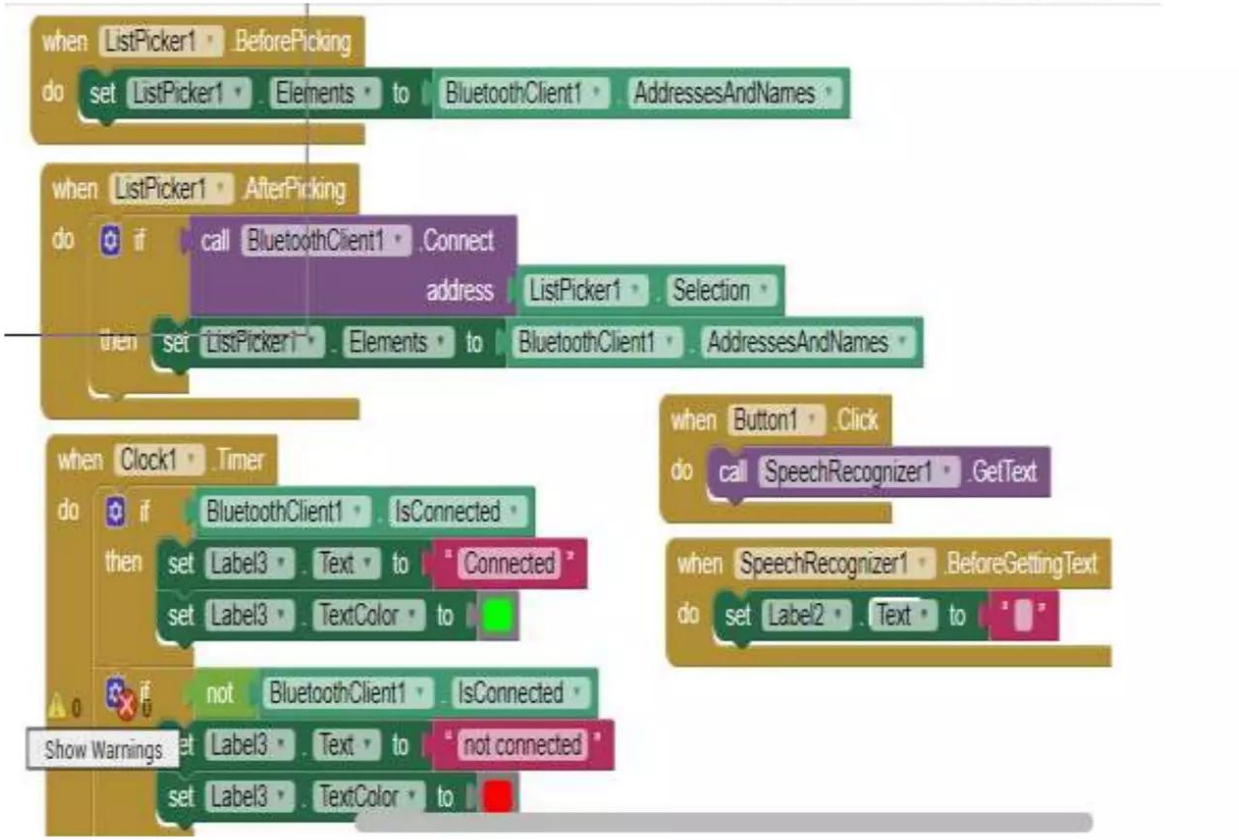


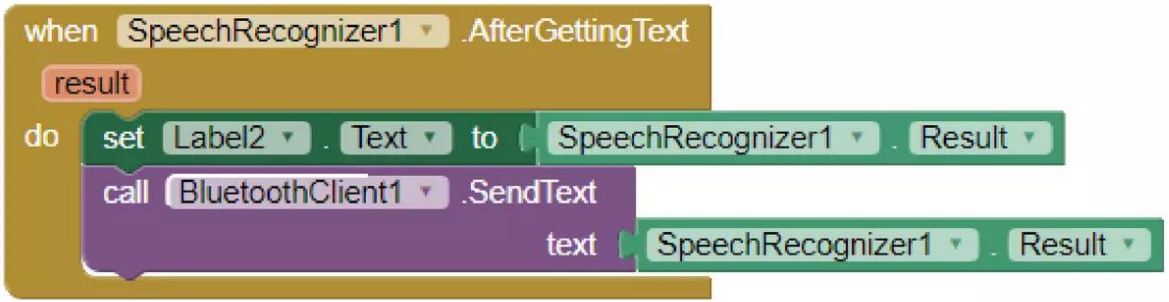
Android Application(Using MIT App Inventor)




ai2.appinventor.mit.edu/?locale=en#5247448770150400
MIT APP INVENTOR
Projects
Connect
Build
Help
My Projects
Gallery
Guide
Report an Issue
English
noorulhuddamemon@gmail.com
SpeechRecognition
Screen1
Add Screen ...
Remove Screen
Designer
Blocks
Blocks
Viewer
Built-in
Control
Logic
Math
Text
Lists
Colors
Variables
Procedures
Screen1
Label3
HorizontalArrangemer
ListPicker1
HorizontalArrangemer
Button1
Label2
SpeechRecognizer1
when ListPicker1 .BeforePicking
do set ListPicker1 . Elements to BluetoothClient1 . AddressesAndNames
when ListPicker1 .AfterPicking
do if call BluetoothClient1 .Connect
address ListPicker1 . Selection
then set ListPicker1 . Elements to BluetoothClient1 . AddressesAndNames
when Clock1 .Timer
do if BluetoothClient1 . IsConnected
then set Label3 . Text to " Connected "
set Label3 . TextColor to
0
0
Show Warnings
if not BluetoothClient1 . IsConnected
then set
when Button1 .Click
do call SpeechRecognizer1 .GetText
when SpeechRecognizer1 .BeforeGettingText
do set Label2 . Text to " "
SpeechRecognitio....apk
voicebot (1).apk
GenYoutube.net_....mp4
GenYoutube.net_....mp4
GenYoutube.net_....mp4
Show all
9:22 PM
9/6/2017

Block Code


when ListPicker1 .BeforePicking
do set ListPicker1 . Elements to BluetoothClient1 . AddressesAndNames
when ListPicker1 .AfterPicking
do if call BluetoothClient1 .Connect
address ListPicker1 . Selection
then set ListPicker1 . Elements to BluetoothClient1 . AddressesAndNames
when Clock1 .Timer
do if BluetoothClient1 . IsConnected
then set Label3 . Text to " Connected "
set Label3 . TextColor to
if not BluetoothClient1 . IsConnected
Show Warnings
Label3 . Text to " not connected "
set Label3 . TextColor to
when Button1 .Click
do call SpeechRecognizer1 .GetText
when SpeechRecognizer1 .BeforeGettingText
do set Label2 . Text to " "


when SpeechRecognizer1 .AfterGettingText
result
do set Label2 . Text to SpeechRecognizer1 . Result
call BluetoothClient1 .SendText
text SpeechRecognizer1 . Result

Software Used..

- Arduino IDE
- MIT App Inventor 2(To create android app)

Programming Languages Used..

- Embedded C/C++
- Android App

Applications

- The robot is useful in places where humans find difficult to reach but human voice reaches. E.g. in a small pipeline, in a fire-situations, in highly toxic areas.
- The robot can be used as a toy.
- It can be used to bring and place small objects.